1 Introduction

Definition 1.1 (Orthonormal set)**.** A set of vectors $\{v_1, \ldots, v_k\}$ in an inner product space V is said to be *orthogonal* if $(v_i, v_j) = 0$ for all $i \neq j$. If, in addition, $||v_i|| = 1$ for all $i = 1, \ldots, k$, then the set is said to be *orthonormal*. In this case, we can write

$$(v_i, v_j) = \delta_{ij},$$

where δ_{ij} is the *Kronecker delta*, defined by

$$\delta_{ij} = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

Example 1.2. The standard basis $\{e_1, \ldots, e_n\}$ of F^n is an orthonormal set with respect to the dot product $v \cdot w = \sum_{i=1}^{n} v_i \overline{w}_i$ on F^n. To prove this, note that we can write $(e_i)_k = \delta_{ik}$, so we have

$$e_i \cdot e_j = \sum_{k=1}^{n} \delta_{ik}\delta_{jk} = \delta_{ij}.$$

Exercise 1.3. Verify that $\{(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}), (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0), (\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}})\}$ is an orthonormal set in F^3 with respect to the dot product.

Exercise 1.4. Show that $\{1, \cos(x), \cos(2x), \cos(3x), \ldots, \sin(x), \sin(2x), \sin(3x), \ldots\}$ is an orthogonal set in $\mathscr{C}([-\pi, \pi])$ (the vector space of continuous real-valued functions on $[-\pi, \pi]$) with respect to the inner product $(f, g) = \int_{-\pi}^{\pi} f(x)g(x)dx$. Normalize each function so that the set is orthonormal.

Orthonormal sets are particularly easy to work with, as illustrated by the next result.

Proposition 1.5. *If $\{e_1, \ldots, e_n\}$ is an orthonormal set of vectors in V and $v = \sum_{i=1}^{n} \alpha_i e_i$ for some scalars $\alpha_i \in F$, then*

$$||v||^2 = \sum_{i=1}^{n} |\alpha_i|^2.$$

Exercise 1.6. Prove Proposition 1.5.

Corollary 1.7. *An orthonormal set of vectors is linearly independent.*

Exercise 1.8. Prove Corollary 1.7.

Now we come to an important inequality.

Proposition 1.9 (Bessel's inequality)**.** *Suppose $\{e_1, \ldots, e_m\}$ is a linearly independent set of vectors in V. Then, for any $v \in V$, $\sum_{i=1}^{m} |(v, e_i)|^2 \leq ||v||^2$.*

Exercise 1.10. Prove Proposition 1.9. (*Hint*: Given $v \in V$, let $u := \sum_{i=1}^{m} (v, e_i) e_i$ and $w := v - u$ so that $v = u + w$. Show that $(u, w) = 0$ and apply the Pythagorean Theorem.

2 Orthonormal Bases

Definition 2.1 (Orthonormal Basis)**.** An *orthonormal basis* of an inner product space V is an orthonormal set of vectors in V which is also a basis of V.

If V is finite-dimensional of dimension $\dim V = n$, then by Corollary 1.7 an orthonormal set of n vectors is an orthonormal basis of V.

Example 2.2.

(a) The standard basis of F^n is an orthonormal basis of F^n.

(b) The orthonormal set in Example 1.3 is also an orthonormal basis of F^3.

Given a basis $\{e_1, \ldots, e_m\}$ of V, for any $v \in V$ there exists unique scalars $\alpha_1, \ldots, \alpha_m$ such that

$$v = \sum_{i=1}^{n} \alpha_i e_i.$$

As we have seen, finding the coefficients α_i involves solving a system of linear equations. However, if $\{e_1, \ldots, e_m\}$ is an orthonormal basis, the next result shows that there is a much easier way to solve for the coefficients: just take $\alpha_k = (v, e_k)$.

Proposition 2.3. *Let $\{e_1, \ldots, e_m\}$ be an orthonormal basis of V and $u, v \in V$. Then*

(a) $v = \sum_{i=1}^{n} (v, e_i) e_i$

(b) $||v||^2 = \sum_{i=1}^{n} |(v, e_i)|^2$

(c) $(u, v) = \sum_{i=1}^{n} (u, e_i)\overline{(v, e_i)}$

Exercise 2.4. Prove Proposition 2.3.

Exercise 2.5. Use Proposition 2.3 to write the vector $(1,2,4,7) \in F^4$ is a linear combination of the orthonormal basis

$$\left\{\left(\frac{1}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}\right),\left(\frac{1}{2},\frac{1}{2},-\frac{1}{2},-\frac{1}{2}\right),\left(\frac{1}{2},-\frac{1}{2},-\frac{1}{2},\frac{1}{2}\right),\left(-\frac{1}{2},\frac{1}{2},-\frac{1}{2},\frac{1}{2}\right)\right\}$$

Exercise 2.6. Let $V = \mathscr{P}_2(\mathbb{R})$ be the vector space of all polynomials with real coefficients of degree at most 2. Show that the standard basis $\{1,x,x^2\}$ is *not* an orthogonal set and hence not an orthonormal basis with respect to the inner product $(p,q) = \int_{-1}^{1} p(x)q(x)dx$.

Now that we understand the usefulness of orthonormal bases, how do we go about finding them? For example, does $\mathscr{P}_n(\mathbb{R})$ with the inner product $(p,q) = \int_{-1}^{1} p(x)q(x)dx$ have an orthonormal basis? The next result will lead to answers to these questions.

The algorithm used in the next proof is called the *Gram-Schmidt procedure*. It gives a method for turning a linearly independent set into an orthonormal set with the same span as the original set.

Proposition 2.7 (Gram-Schmidt procedure)**.** *Suppose $\{v_1,\dots,v_m\}$ is a linearly independent set of vectors in V. Let $f_1 = v_1$. For $k = 2,\dots,m$, define f_k inductively by*

$$f_k = v_k - \sum_{j=1}^{k-1} \frac{(v_k, f_j)}{||f_j||^2} f_j.$$

For each $k = 1,\dots,m$, let $e_k = \frac{f_k}{||f_k||}$. Then $\{e_1,\dots,e_m\}$ is an orthonormal set of vectors in V such that

$$span\{v_1,\dots,v_k\} = span\{e_1,\dots,e_k\}$$

for each $k = 1,\dots,m$.

Exercise 2.8. Prove Proposition 2.7.

Exercise 2.9. Use the Gram-Schmidt procedure to convert $\{1, x, x^2\}$ into an orthonormal basis of $\mathscr{P}_2(\mathbb{R})$.

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Corollary 2.10. *Every finite-dimensional inner product space has an orthonormal basis.*

Corollary 2.11. *Let V be a finite-dimensional inner product space. Then every orthonormal set of vectors in V can be extended to an orthonormla basis of V.*